HashBrown v1.1.0

MG1655

Assembly report · 2026-10-26

✓ High-quality draft (MIMAG)

GENOME SIZE	CONTIGS	N50	COVERAGE
4,640,169 bp	1 %GC 50.79	4,640,169 bp	100x / 59x long / short reads

SPECIES	MLST	CHECKM2	CONFINDR
Escherichia coli str. K-12... ANI 100.00%	ST10 ecoli_achtman_4	100.0% contam. 0.14%	Clean (1 SNV) Escherichia

QC criteria

METRIC	MEASURED	THRESHOLD	STATUS
CheckM2 completeness	100.0%	> 90% (MIMAG high-quality)	✓ Pass
CheckM2 contamination	0.14%	< 5% (MIMAG)	✓ Pass
rRNA genes (5S, 16S, 23S)	8x 5S, 7x 16S, 7x 23S	≥ 1 of each (MIMAG)	✓ Pass
tRNA isotypes	89 tRNAs, 24 isotypes	≥ 18 (MIMAG)	✓ Pass
Long-read coverage	100x	≥ 30x	✓ Pass
Short-read coverage	59x	≥ 30x	✓ Pass
ANI to closest genome	100.00%	≥ 95% (same species)	✓ Pass
ConFindr (intra-species)	Clean (1 contaminating SNV)	Clean	✓ Pass

Thresholds follow the MIMAG high-quality draft standard (Bowers et al. 2017, doi:10.1038/nbt.3893) plus pipeline read-coverage minimums. A metric whose module did not run is reported as not assessed, never as a pass.

Read QC - Long reads

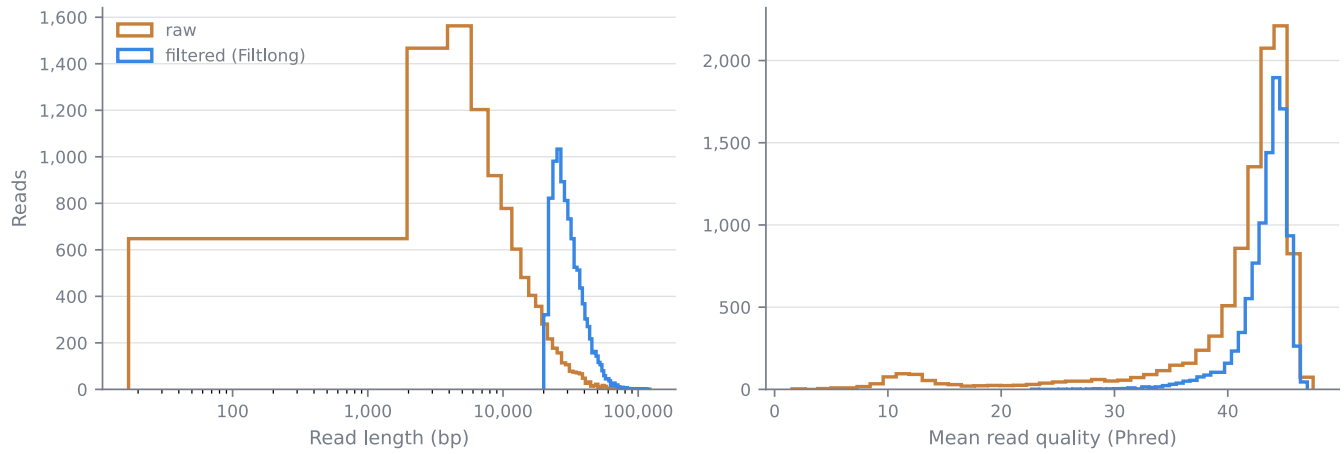
STAGE	READS	BASES	N50	MEAN Q	%GC	COVERAGE	% OF RAW
raw	184,984	2,077,119,077	16,462	17.6	50.7	441x	100.0%
filtered (Filtlong)	14,060	464,018,169	33,230	23.4	50.8	100x	22.3%

Coverage is the mapped mean depth of that stage's reads on the final assembly (mosdepth); a ~ marks a base-count estimate where the stage was not mapped.

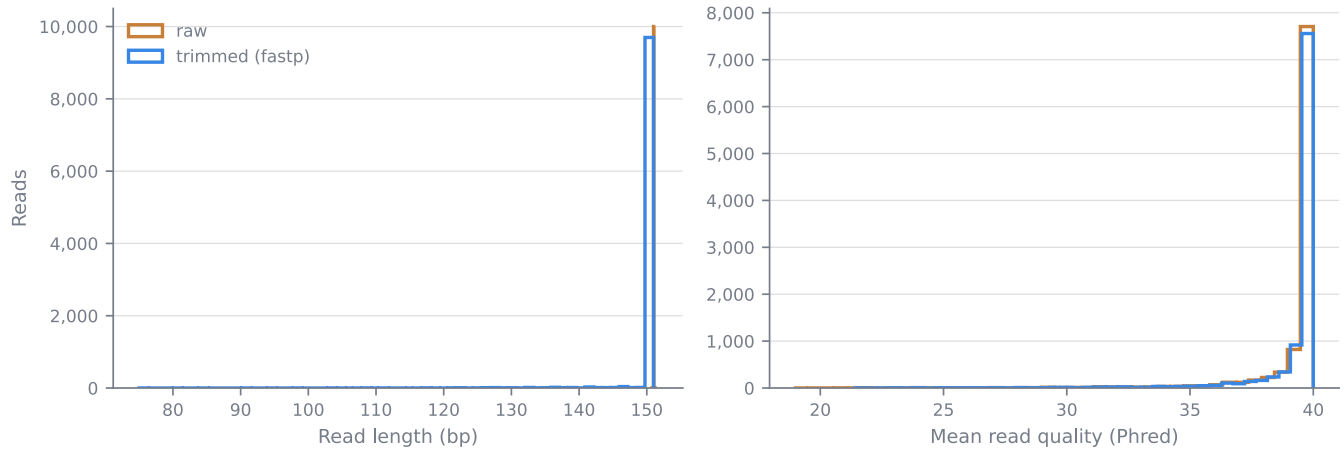
Read QC - Short reads

STAGE	READS	BASES	N50	MEAN Q	%GC	COVERAGE	% OF RAW
raw	1,860,000	280,860,000	151	29.9	50.4	59x	100.0%
trimmed (fastp)	1,848,624	278,203,509	151	30.5	50.4	59x	99.1%

Coverage is the mapped mean depth of that stage's reads on the final assembly (mosdepth); a ~ marks a base-count estimate where the stage was not mapped.



Long reads: length and quality by stage (first 10000 reads per stage)



Short reads: length and quality by stage (first 10000 reads per stage)

Assembly - stage by stage

STAGE	CONTIGS	SIZE (BP)	N50	%GC	QV
Assembled (long reads)	1	4,640,169	4,640,169	50.79	50.0825
Polished - Medaka	1	4,640,169	4,640,169	50.79	50.0825
Polished - short reads	1	4,640,169	4,640,169	50.79	50.2776
Reoriented (dnaapler)	1	4,640,169	4,640,169	50.79	50.2776

QV: Merqury consensus quality from the short-read k-mers (Q30 ≈ 1 error/kb, Q50 ≈ 1 error/100 kb; higher is better).

Species confirmation

CLOSEST GENOME (MASH)	MASH DISTANCE	ORGANISM	ANI
GCF_000005845.2_ASM584v2_genomic.fna.gz	0	*Escherichia coli str. K-12 substr. MG1655*	100.00%

Same species: ANI ≥ 95% against the closest genome.

Completeness & sanity

METRIC	VALUE
CheckM2 completeness	100.0%
CheckM2 contamination	0.14%
BUSCO (enterobacteriaceae_odb12.2)	C:99.7%[S:99.5%,D:0.1%],F:0.2%,M:0.1%,n:874
rRNA genes (barrnap)	8x 5S, 7x 16S, 7x 23S
tRNAs (tRNAscan-SE)	89 tRNAs, 24 isotypes

Typing

MLST SCHEME	ST	ALLELES
ecoli_achtman_4	10	adk(10) fumC(11) gyrB(4) icd(8) mdh(8) purA(8) recA(2)

Plasmids (MOB-suite)

No plasmids detected.

AMR & virulence genes (89)

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
emrD	AMRFinderPlus	100.00	100.00	contig_1	EFFLUX / EFFLUX
arsC	AMRFinderPlus	94.33	100.00	contig_1	ARSENIC / ARSENATE
arsR	AMRFinderPlus	100.00	100.00	contig_1	ARSENIC / ARSENIC
acrF	AMRFinderPlus	100.00	100.00	contig_1	EFFLUX / EFFLUX
asr	AMRFinderPlus	100.00	100.00	contig_1	NA / NA

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
ariR	AMRFinderPlus	100.00	100.00	contig_1	NA / NA
emrE	AMRFinderPlus	100.00	100.00	contig_1	EFFLUX / EFFLUX
mdtM	AMRFinderPlus	100.00	100.00	contig_1	EFFLUX / EFFLUX
blaEC	AMRFinderPlus	100.00	100.00	contig_1	BETA-LACTAM / BETA-LACTAM
fieF	AMRFinderPlus	100.00	100.00	contig_1	NA / NA
gspC	VFDB	95.41	99.90	contig_1	(gspC) general secretion pathway protein C [T2SS (VF0333) - Effector delivery system (VFC0086)] [Shigella dysenteriae Sd197]
gspL	VFDB	96.89	81.85	contig_1	(gspL) general secretion pathway protein L [T2SS (VF0333) - Effector delivery system (VFC0086)] [Shigella dysenteriae Sd197]
gspM	VFDB	97.95	100.00	contig_1	(gspM) general secretion pathway protein M [T2SS (VF0333) - Effector delivery system (VFC0086)] [Shigella dysenteriae Sd197]
rpoS	VFDB	92.75	99.80	contig_1	(rpoS) RNA polymerase sigma factor RpoS [RpoS (VF0112) - Regulation (VFC0301)] [Salmonella enterica subsp. enterica serovar Typhimurium str. LT2]
gtrB	VFDB	80.66	97.31	contig_1	(gtrB) bactoprenol glucosyl transferase [LPS (VF0124) - Immune modulation (VFC0258)] [Shigella flexneri 2a str. 301]
gtrA	VFDB	88.70	100.00	contig_1	(gtrA) bactoprenol-linked glucose translocase/ flippase [LPS (VF0124) - Immune modulation (VFC0258)] [Shigella flexneri 2a str. 301]
rcsB	VFDB	84.48	100.00	contig_1	(rcsB) transcriptional regulator RcsB [RcsAB (VF0571) - Regulation (VFC0301)] [Klebsiella pneumoniae subsp. pneumoniae NTUH-K2044]
gndA	VFDB	84.22	100.00	contig_1	(gndA) NADP-dependent phosphogluconate dehydrogenase [Capsule (VF0560) - Immune modulation (VFC0258)] [Klebsiella pneumoniae subsp. pneumoniae NTUH-K2044]
fliQ	VFDB	82.33	98.52	contig_1	(fliQ) flagellar biosynthesis protein FliQ [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
fliP	VFDB	82.40	91.06	contig_1	(fliP) flagellar type III secretion system pore protein FliP [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
fliN	VFDB	84.99	99.76	contig_1	(fliN) flagellar motor switch protein FliN [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
fliM	VFDB	84.28	100.00	contig_1	(fliM) flagellar motor switch protein FliM [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
fliI	VFDB	82.26	99.20	contig_1	(fliI) flagellar protein export ATPase FliI [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
fliG	VFDB	86.87	99.40	contig_1	(fliG) flagellar motor switch protein FliG [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
fliE	VFDB	81.79	97.78	contig_1	(fliE) flagellar hook-basal body complex protein FliE [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
espR4	VFDB	83.96	99.47	contig_1	(espR4) Type III secretion system effector espR4 [TTSS secreted effectors (VF1111) - Effector delivery system (VFC0086)] [Escherichia coli O55:H7 str. CB9615]
flhC	VFDB	82.24	99.83	contig_1	(flhC) flagellar transcriptional regulator FlhC [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
motA	VFDB	82.32	100.00	contig_1	(motA) flagellar motor stator protein MotA [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
motB	VFDB	80.48	84.84	contig_1	(motB) flagellar motor protein MotB [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
cheW	VFDB	83.03	97.02	contig_1	(cheW) chemotaxis protein CheW [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
cheB	VFDB	82.06	99.81	contig_1	(cheB) protein-glutamate methylesterase/protein-glutamine glutaminase [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
cheY	VFDB	83.85	100.00	contig_1	(cheY) chemotaxis response regulator CheY [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
cheZ	VFDB	80.16	100.00	contig_1	(cheZ) protein phosphatase CheZ [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
flhA	VFDB	82.11	99.76	contig_1	(flhA) flagellar biosynthesis protein FlhA [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
espL1	VFDB	97.58	100.00	contig_1	(espL1) Type III secretion system effector espL1 [TTSS secreted effectors (VF1111) - Effector delivery system (VFC0086)] [Escherichia coli O55:H7 str. CB9615]
espR1	VFDB	93.94	83.73	contig_1	(espR1) Type III secretion system effector espR1 [TTSS secreted effectors (VF1111) - Effector delivery system (VFC0086)] [Escherichia coli O55:H7 str. CB9615]
phoP	VFDB	82.09	99.26	contig_1	(phoP) response regulator in two-component regulatory system with PhoQ [PhoPQ (VF0111) - Regulation (VFC0301)] [Salmonella enterica subsp. enterica serovar Typhimurium str. LT2]
flgI	VFDB	80.69	95.54	contig_1	(flgI) flagellar basal body P-ring protein FlgI [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
flgH	VFDB	81.31	99.71	contig_1	(flgH) flagellar basal body L-ring protein FlgH [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
flgG	VFDB	84.93	100.00	contig_1	(flgG) flagellar basal-body rod protein FlgG [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
flgC	VFDB	82.72	100.00	contig_1	(flgC) flagellar basal body rod protein FlgC [Flagella (VF1400) - Motility (VFC0204)] [Enterobacter cloacae strain M12X01451]
csgC	VFDB	98.80	100.00	contig_1	(csgC) curli assembly protein CsgC [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
csgA	VFDB	91.72	99.35	contig_1	(csgA) curlin major subunit CsgA [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
csgB	VFDB	99.78	100.00	contig_1	(csgB) curlin minor subunit CsgB [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
cgsD	VFDB	99.08	100.00	contig_1	(cgsD) transcriptional regulator CsgD [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
cgsE	VFDB	98.72	100.00	contig_1	(cgsE) curli production assembly/transport protein CsgE [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
cgsF	VFDB	98.08	100.00	contig_1	(cgsF) curli production assembly/transport protein CsgF [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
cgsG	VFDB	97.12	100.00	contig_1	(cgsG) curli production assembly/transport protein CsgG [Curli fibers (VF1138) - Adherence (VFC0001)] [Escherichia coli O25b:H4-ST131]
ompA	VFDB	98.08	100.00	contig_1	(ompA) outer membrane protein A [OmpA (VF0236) - Invasion (VFC0083)] [Escherichia coli O18:K1:H7 str. RS218]
fur	VFDB	87.64	96.47	contig_1	(fur) ferric iron uptake transcriptional regulator [Fur (VF0113) - Regulation (VFC0301)] [Salmonella enterica subsp. enterica serovar Typhimurium str. LT2]
entA	VFDB	95.72	100.00	contig_1	(entA) 23-dihydro-23-dihydroxybenzoate dehydrogenase EntA [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entB	VFDB	98.14	100.00	contig_1	(entB) isochorismatase [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entE	VFDB	95.47	100.00	contig_1	(entE) 23-dihydroxybenzoate-AMP ligase component of enterobactin synthase multienzyme complex [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entC	VFDB	97.87	100.00	contig_1	(entC) isochorismate synthase 1 [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fepB	VFDB	97.49	100.00	contig_1	(fepB) ferrienterobactin ABC transporter periplasmic binding protein [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entS	VFDB	95.44	100.00	contig_1	(entS) enterobactin exporter iron-regulated [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fepD	VFDB	96.02	100.00	contig_1	(fepD) ferrienterobactin ABC transporter permease [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fepG	VFDB	94.06	100.00	contig_1	(fepG) iron-enterobactin ABC transporter permease [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fepC	VFDB	97.30	100.00	contig_1	(fepC) ferrienterobactin ABC transporter ATPase [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
fepE	VFDB	94.53	100.00	contig_1	(fepE) LPS O-antigen length regulator [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entF	VFDB	95.69	99.28	contig_1	(entF) enterobactin synthase multienzyme complex component ATP-dependent [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fes	VFDB	96.67	100.00	contig_1	(fes) enterobactin/ferric enterobactin esterase [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
fepA	VFDB	96.70	100.00	contig_1	(fepA) ferrienterobactin outer membrane transporter [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
entD	VFDB	95.01	100.00	contig_1	(entD) phosphopantetheinyl transferase component of enterobactin synthase multienzyme complex [Enterobactin (VF0228) - Nutritional/Metabolic factor (VFC0272)] [Escherichia coli CFT073]
ibeB	VFDB	96.53	99.93	contig_1	(ibeB) Cu(+)/Ag(+) efflux RND transporter outer membrane channel CusC [Ibes (VF0237) - Invasion (VFC0083)] [Escherichia coli O45:K1:H7 str. S88]
allB	VFDB	82.86	99.78	contig_1	(allB) allantoinase [Allantion utilization (VF0572) - Nutritional/Metabolic factor (VFC0272)] [Klebsiella pneumoniae subsp. pneumoniae NTUH-K2044]
acrB	VFDB	83.88	99.84	contig_1	(acrB) acriflavine resistance protein B [AcrAB (VF0568) - Antimicrobial activity/Competitive advantage (VFC0325)] [Klebsiella pneumoniae subsp. pneumoniae NTUH-K2044]
ykgK/ ecpR	VFDB	97.24	100.00	contig_1	(ykgK/ecpR) regulator protein EcpR [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]
yagZ/ ecpA	VFDB	99.15	100.00	contig_1	(yagZ/ecpA) E. coli common pilus structural subunit EcpA [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]
yagY/ ecpB	VFDB	98.06	100.00	contig_1	(yagY/ecpB) E. coli common pilus chaperone EcpB [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]
yagX/ ecpC	VFDB	98.77	100.00	contig_1	(yagX/ecpC) E. coli common pilus usher EcpC [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
yagW/ ecpD	VFDB	99.09	100.00	contig_1	(yagW/ecpD) polymerized tip adhesin of ECP fibers [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]
yagV/ ecpE	VFDB	97.04	99.72	contig_1	(yagV/ecpE) E. coli common pilus chaperone EcpE [ECP (VF0404) - Adherence (VFC0001)] [Escherichia coli O157:H7 str. EDL933]
espY1	VFDB	88.58	99.87	contig_1	(espY1) Type III secretion system effector EspY1 [TTSS secreted effectors (VF1110) - Effector delivery system (VFC0086)] [Escherichia coli O157:H7 str. EDL933]
fimH	VFDB	97.23	100.00	contig_1	(fimH) FimH protein precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimG	VFDB	98.02	100.00	contig_1	(fimG) FimG protein precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimF	VFDB	97.36	100.00	contig_1	(fimF) FimF protein precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimD	VFDB	98.22	100.00	contig_1	(fimD) Outer membrane usher protein fimD precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimC	VFDB	98.35	100.00	contig_1	(fimC) Chaperone protein fimC precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimI	VFDB	98.33	100.00	contig_1	(fimI) Fimbrin-like protein fimI precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimA	VFDB	91.98	100.00	contig_1	(fimA) Type-1 fimbrial protein A chain precursor [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimE	VFDB	98.66	100.00	contig_1	(fimE) Type 1 fimbriae Regulatory protein fimE [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
fimB	VFDB	98.18	100.00	contig_1	(fimB) Type 1 fimbriae Regulatory protein fimB [Type 1 fimbriae (VF0221) - Adherence (VFC0001)] [Escherichia coli CFT073]
pmrA	VFDB	81.02	100.00	contig_1	(pmrA) response regulator PmrA [PmrAB (VF1355) - Regulation (VFC0301)] [Salmonella enterica subsp. enterica serovar Typhimurium str. LT2]

GENE	SOURCE	%IDENTITY	%COVERAGE	CONTIG	CLASS / PRODUCT
espX5	VFDB	97.53	100.00	contig_1	(espX5) Type III secretion system effector EspX5 [TTSS secreted effectors (VF1111) - Effector delivery system (VFC0086)] [Escherichia coli O55:H7 str. CB9615]
espX4	VFDB	93.36	99.94	contig_1	(espX4) Type III secretion system effector EspX4 [TTSS secreted effectors (VF1110) - Effector delivery system (VFC0086)] [Escherichia coli O157:H7 str. EDL933]
espL4	VFDB	97.30	100.00	contig_1	(espL4) Type III secretion system effector EspL4 [TTSS secreted effectors (VF1110) - Effector delivery system (VFC0086)] [Escherichia coli O157:H7 str. EDL933]
ibeC	VFDB	97.64	100.00	contig_1	(ibeC) phosphoethanolamine transferase CptA [Ibes (VF0237) - Invasion (VFC0083)] [Escherichia coli O45:K1:H7 str. S88]
aslA	VFDB	93.56	100.00	contig_1	(aslA) putative arylsulfatase [AslA (VF0238) - Invasion (VFC0083)] [Escherichia coli O18:K1:H7 str. RS218]

AMRFinderPlus: resistance genes and point mutations. VFDB: virulence factors. PlasmidFinder: plasmid replicons.

Annotation

Bakta: 88 tRNAs, 22 rRNAs, 231 ncRNAs, 2 CRISPR arrays, 4288 CDSs

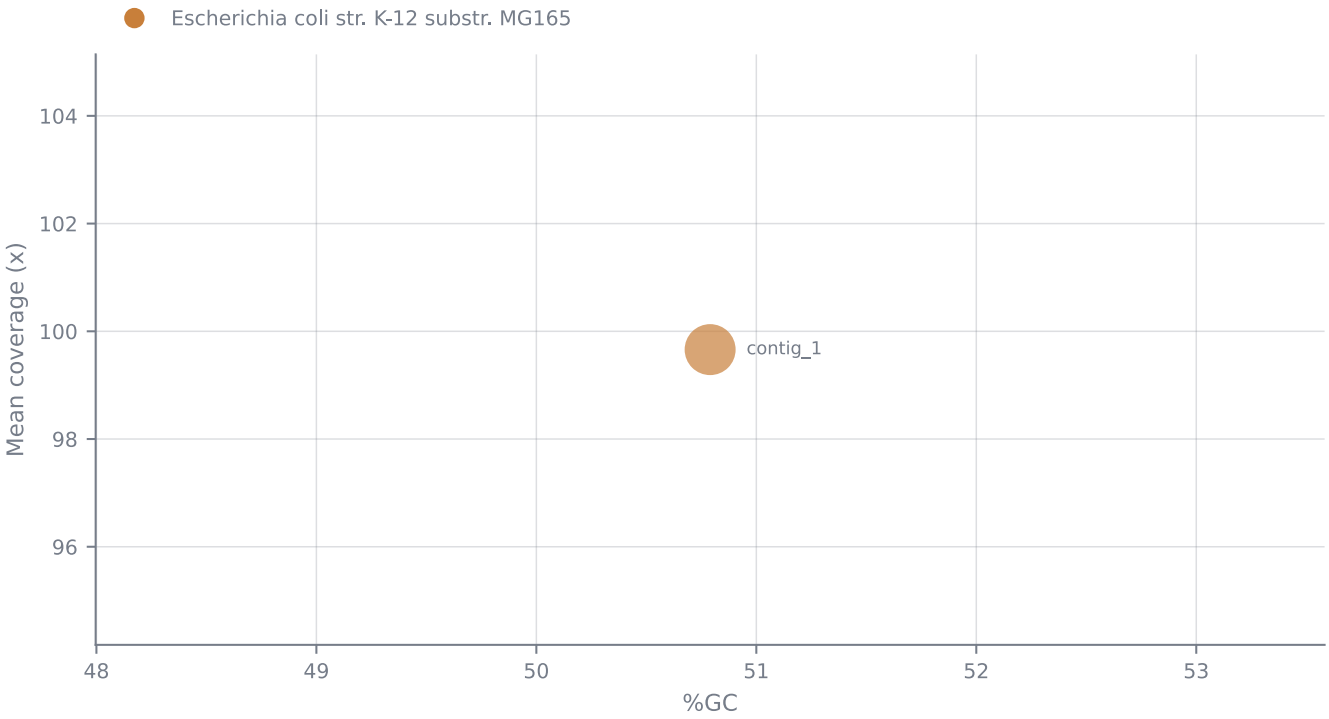
Read-based genome profile (GenomeScope2)

PROPERTY	VALUE
Estimated haploid genome size	4,647,372 bp - 4,649,296 bp
Read error rate	0.0940699% - 0.0940699%

From the short-read 21-mers, independent of the assembly: a size estimate far from the assembly size, or non-zero heterozygosity, points at contamination or a mixed culture.

Contamination

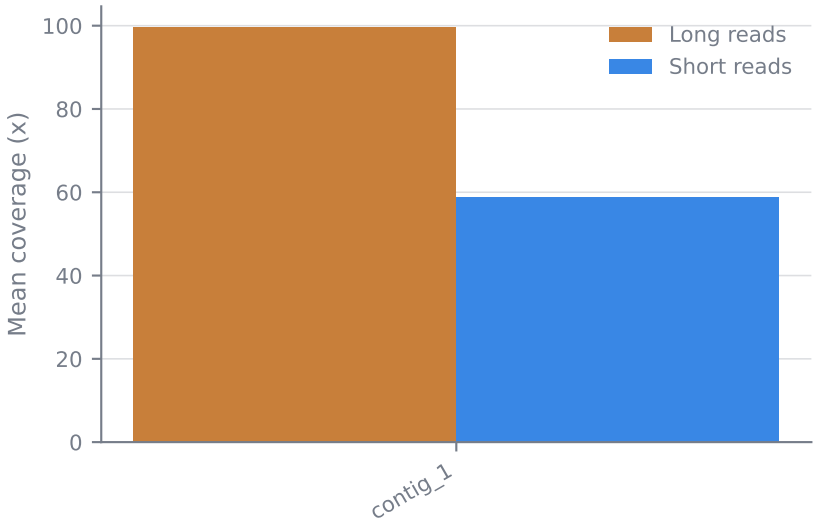
CONFINDR VERDICT	GENUS	CONTAMINATING SNVS
Clean	Escherichia	1



Blob plot: every contig by %GC and coverage (dot area ~ length, color = closest RefSeq genome). Contigs far from the main blob, or colored as another species, are contamination candidates.

Coverage per contig (mosdepth)

CONTIG	LENGTH (BP)	LONG-READ MEAN	SHORT-READ MEAN
contig_1	4,640,169	100x	59x
total	**4,640,169**	**100x**	**59x**



Detailed reports

- QUAST report

- Assembly graph (Bandage)
- Annotation (Bakta, GFF3)
- pycoQC - long reads, raw
- pycoQC - long reads, filtered (Filtlong)
- pycoQC - short reads, raw
- pycoQC - short reads, trimmed (fastp)

Traceability

ITEM	VALUE
HashBrown	v1.1.0
Report generated	2026-10-26 19:00
Command line	`hashbrown -l /media/8tb/analyses/hashbrown_benchmark/example_full/reads_long -s /media/8tb/analyses/hashbrown_benchmark/example_full/reads_short -o /media/8tb/analyses/hashbrown_benchmark/example_full/run --filter-long --trim-short --size 4640000 -t 40 -p 1`

SOFTWARE VERSIONS

SOFTWARE	VERSION
Python	Python 3.12.15
Flye	2.9.6-b1802
Shasta	Shasta Release 0.14.0
Medaka	medaka 2.2.2
Porechop	0.2.4
Filtlong	Filtlong v0.3.1
Fastp	fastp 1.4.0
BWA	Version: 0.7.19-r1273
Polypolish	Polypolish 0.7.1
pyPOLCA	pypolca, version 0.5.0
SAMtools	samtools 1.24
Minimap2	2.31-r1302
Seqtk	Version: 1.5-r133
QUAST	QUAST v5.3.0
mosdepth	mosdepth 0.3.14
SeqKit	seqkit v2.14.0

SOFTWARE	VERSION
MUMmer (nucmer)	4.0.1
Bandage	Version: 0.9.0

Tables as TSV: `tables/`